Privacy – Enhanced Synthetic Data Generation for Autonomous Vehicle Training

R26-CS-010

Project Proposal Report

APEKSHA M.K.S.R – IT22110220

B.Sc. (Hons) Degree in Information Technology Specialized in Cyber Security

Department of Computer Systems Engineering

Sri Lanka Institute of Information Technology Sri Lanka

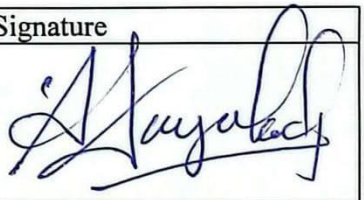
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
APEKSHA M.K.S.R	**IT22110220**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Prof.Anuradha Jayakody	**16/03/2026**	

Name	Date	Signature
Mr. Dinith Primal	**16/03/2026**	

ABSTRACT

Autonomous Vehicles (AVs) make use of large annotated data sets for training perception models for various tasks like object detection, pedestrian detection etc. There are various different privacy issues that are encountered with the use of AVs for generating the data required for driving. For example, it has been noticed that the AVs may reveal various kinds of data like faces, license plates etc. There is a requirement for the development of data generation techniques that help to resolve the privacy issues due to the implementation of data protection policies like GDPR.

However, recent research works were able to focus on the utilization of synthetic datasets to minimize the dependence of AVs on the data gathered from the real world. There is a significant research problem that seeks to find out if synthetic datasets that are enhanced to remove privacy concerns can be utilized for training the perception models of AVs.

The goal of the research is to propose a Privacy Enhanced Synthetic Data Generation Framework for the Training of Autonomous Vehicle perception models, wherein synthetic driving environments will be generated utilizing the simulated platforms that will incorporate the privacy preservation techniques in the data generation process.

The technologies to be used for the proposed research include CARLA simulation platform, PyTorch deep learning framework, YOLO-based perception models, and automated labeling techniques. The Performance metrics for the trained models will measured based on mAP, precision and recall metrics to assess the impact of privacy constraints.

The expected outcome of this research is that it would quantitatively evaluate the privacy-utility trade-off, which would also help in understanding the impact of synthetic data with privacy – preserving techniques on the accuracy of perception for AVs, thereby creating privacy-compliant AI that is ethical solutions for intelligent transportation systems (ITS) applications.

The research proposed is also in alignment with Sustainable Development Goals:SDG 9: Industry, Innovation and Infrastructure; SDG 11: Sustainable cities and communities; and SDG 16: Peace, Justice and Strong Institutions.

Keywords: Autonomous Vehicles, Synthetic Data, Privacy – Preserving AI, Differential Privacy, Perception Models.

ACKNOWLEDGEMENT

I would like to extend my sincere and heartfelt appreciation to my research supervisor Prof. Anuradha Jayakody who provided valuable guidance and constructive feedback throughout the development of our research project. The expertise of our research supervisor in artificial intelligence and intelligence systems significantly contributed to shaping our research project.

I also would like to thank my group members and colleagues for their cooperation and discussions during the early stages of this project. I acknowledge the contributions of the research community whose published work has provided valuable insights for this proposal.

Lastly, I want to express our heartfelt thanks to my families for their encouragement, patience and unflinching support during the completion of our research project.

TABLE OF CONTENT

Abstract.....3

Acknowledgement.....4

Table of Content 5

List of tables 7

List of figures.....7

List of Abbreviations 8

List of appendices.....8

1. Introduction 9

1.1 Background and Context 9

1.2 Literature Review 10

1.3 Research Gap 11

2. Objective 12

2.1 Main Objective.....12

2.2 Specific Objectives 12

3. Methodology 13

3.1Research Approach.....13

3.2 Technical methods and algorithms.....13

3.3 Tools and Platforms.....14

3.4 Data collection and Ethical Consideration.....14

3.5 Validation strategy and performance metrics.....15

4. High - level system architecture.....16

4.1 Overview of proposed solution and its components.....16

4.2 Main Components of the System.....17

4.3 Data flow explanation.....18

4.4 Scalability and security considerations.....21

5. User requirements.....22

5.1 Stakeholder Identification.....22

5.2 Functional Requirements.....22

5.3 Non - Functional Requirements……23
6. Commercialization Plan……...24
6.1 Target market and customer persona……...24
6.2 Value proposition……24
6.3 Revenue model……25
6.4 Competitive Advantage……...25
6.5 Cost estimation……26
6.6 Pricing strategy……....26
6.7 Intellectual Property Considerations……26
7. Budget and Justification……27
8. Work Breakdown Structure……..28
8.1Task Breakdown per member……..28
8.2 Timeline Alignment……..30
8.3 Balanced Workload Distribution……30
9. Gantt Chart……....32
10. Reference list……33
11. Appendices……34
11.1 Appendix A Survey Instruments……..34
11.2 Appendix B Detailed System Diagrams……...34
11.3 Appendix C Preliminary User Interface Mockups……35
11.4 Appendix D Risk Analysis……35
11.5 Appendix E Ethical Documents……....36

LIST OF TABLES

Table 3.3: Tools and Platforms

Table 7: Budget Estimation for Research Project

Table 8.1: Work Breakdown Structure (WBS)

Table 8.2: Timeline Alignment

Table 8.3: Balanced Workload Distribution

Table 11.1: Appendix A – Survey Instruments

Table 11.2: Appendix B – Detailed System Diagrams

Table 11.3: Appendix C – Preliminary User Interface Mockups

Table 11.4: Appendix D – Risk Analysis

Table 11.5: Appendix E – Ethical Documents

LIST OF FIGURES

Figure 4.3: Data Flow Explanation

Figure 9: Gantt Chart

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
AV	Autonomous Vehicle
CARLA	Car Learning to Act
CV	Computer Vision
DP	Differential Privacy
GDPR	General Data Protection Act
LiDAR	Light Detection and Ranging
ML	Machine Learning
SDG	Sustainable Development Goal
WBS	Work Breakdown Structure

LIST OF APPENDICES

Appendix A – Survey Instruments

Appendix B – Detailed System Diagrams

Appendix C – Preliminary User Interface Mockups

Appendix D – Risk Analysis

Appendix E – Ethical Documents

1. INTRODUCTION

1.1 Background and Context

The use of advanced artificial intelligence systems is necessary for the development of autonomous vehicles. This artificial intelligence system requires extensive data obtained from the real world, including images, videos and sensor data obtained from cameras, LiDAR and radar systems. This indicates that the accuracy of autonomous vehicle perception systems is greatly affected by the availability of extensive annotated data.

The use of real-world driving data to develop autonomous vehicles faces a number of privacy issues. There are number of real world scenes where a lot of personal data is involved. There are faces of pedestrians, license plates of other vehicles, and location – specific data. Organizations have to adhere to data privacy laws such as the General Data Protection Regulation (GDPR) to maintain the security of the data.

The autonomous vehicle market is expanding rapidly all over the globe. According to market reports, the global autonomous vehicle market may reach 500 billion dollars by 2030. This necessitates the need to gather a large amount of data to train artificial intelligence. Traditional methods of data collection are expensive and may raise ethical concerns.

To address all these problems, attempts have been made to leverage synthetic data sets created with the aid of simulation tools like CARLA. Though the application of synthetic data sets can minimize the risk of privacy violations to a certain extent, it is not clear whether privacy – preserving synthetic data sets can be used for training AV perception models accurately.

The general objective of conducting this research is to design privacy – enhanced synthetic data set generation mechanism that can potentially leverage both simulated data set generation and privacy-preserving techniques. The mechanism can leverage synthetic driving data sets to train AV perception models and evaluate the AV perception model.

The proposed research is related to one of the AIMS (Autonomous Intelligent Machines and Systems) research clusters and can contribute to developing intelligent and privacy-aware AI systems. Furthermore, this research can help achieve various United Nations Sustainable Development Goals, including SDG 9, SDG 11, and SDG 16.

1.2 Literature Review

Recent research has emphasized synthetic data generation and privacy-preserving machine learning methods to deal with issues related to data collection.

Some research has also considered simulation platforms to develop synthetic data. For example, **Dosovitskiy et al**. presented CARLA as an open urban driving simulator to train and validate autonomous driving systems. The simulator can provide realistic scenarios with varying weather conditions.

Some research has also considered synthetic data generation to train deep leaning – based perception systems. For example, **Ros et al.** showed that synthetic data can significantly improve object detection system performance using deep learning methods. Synthetic data can provide diverse scenarios, which are hard to collect in real – world scenarios.

Privacy – preserving machine learning has also attracted significant research interest. Researchers have introduced differential privacy methods to avoid data breaches during model training. For example, **Abadi et al.** introduced differentially private stochastic gradient descent to maintain privacy during deep learning system training.

For instance, in the context of AV, perception models such as YOLO and Faster R-CNN are commonly used for real-time object detection. These models need significant data sets to ensure high accuracy in object detection.

While significant advancements have been achieved in the context of synthetic data sets and synthetic AI, the existing literature has mainly focused on the development of data sets instead of evaluating the impact of data privacy on the performance and utility of AV perception models.

Hence, there is still a need to conduct further research to evaluate the impact of data sets with privacy constraints on the performance and utility of AV perception models.

1.3 Research Gap

Although there is significant research on synthetic data generation and privacy preservation techniques, there are various limitations in this area of research that are yet to be addressed.

One of the limitations is that most of the existing research on synthetic data generation is focused on creating a synthetic dataset that is similar in appearance to a real dataset. However, there is a lack of research on how the privacy preservation techniques affect the performance of the perception models of an autonomous vehicle.

Another limitation of the existing research is that most of the privacy preservation techniques are applied after the dataset generation as a post – processing step. This may affect the quality of the generated synthetic data.

Most of the existing research lacks a quantitative analysis of the accuracy of the models.

Therefore, there is a research gap with regard to evaluating the utility of such privacy-respecting synthetic datasets for training autonomous vehicle perception models.

With regard to this research gap, this research aims to create privacy-preserving synthetic datasets and analyze the utility of such datasets with regard to training perception models.

2. OBJECTIVE

2.1 Main Objective

To determine the effectiveness of privacy enhanced synthetic datasets for training autonomous vehicle perception models by measuring the privacy utility trade-off using quantitative metrics.

2.2 Specific Objectives

1. To create synthetic autonomous driving datasets with the help of simulation platforms.
2. To ensure the preservation of privacy while creating synthetic datasets.
3. To train autonomous vehicle perception models with the synthetic dataset developed in the previous objective.
4. To measure the effectiveness of the model using metrics such as mAP (Mean Average Precision).
5. To examine the trade-off between privacy and model accuracy.

3. METHODOLOGY

3.1 Research Approach

This research applies an experimental research methodology to assess the effectiveness of using privacy-enhanced synthetic datasets in training perception models for autonomous vehicles.

The first step in this experimental research methodology is to create synthetic driving datasets using a simulation environment such as CARLA. At this point in the research methodology, there are various privacy settings that can be tested and evaluated in order to access their impact on the effectiveness of the perception models that are to be trained.

Once the datasets are created, deep learning-based perception models are to be trained using these synthetic datasets.

Before training the perception models using synthetic datasets, a baseline is to be established using real world datasets such as the KITTI dataset.

Besides establishing this baseline using real-world data such as KITTI, the training parameters of the model training pipeline would also be systematically set by configuring parameters such as batch size, learning rate, number of epochs, etc. The models that were trained using synthetic data would then be compared to the non – private baseline.

Next, the perception models will be trained using the synthetic dataset with the help of a deep learning framework like PyTorch. During training, the perception models will be able to recognize and identify various objects commonly encountered in driving scenarios like cars, pedestrians, signs and roads.

Finally, the experiment's outcomes will be analyzed to understand the privacy-utility trade-off. In this analysis, the impact of privacy constraints imposed during dataset generation on the performance of the perception model for an autonomous vehicle is measured.

The results will help determine whether privacy – preserving synthetic datasets can still provide sufficient utility for training effective AV perception systems.

3.2 Technical Methods and Algorithms

The proposed system would utilize deep learning-based object detection algorithms to train and evaluate the autonomous vehicle perception models. Object detection algorithms are commonly employed in computer vision applications to detect and recognize objects within an image.

The first object detection algorithm employed in this research is the YOLO object detection model. YOLO is an object detection algorithm that can detect multiple objects within an image simultaneously. This algorithm works by processing the entire image in one go and then predicting the location of objects such as vehicles, pedestrians and traffic signs. This

algorithm is employed in autonomous vehicle applications owing to its high speed and accuracy.

The system will also use Convolutional Neural Networks (CNNs), which are a class of deep learning model that are particularly suitable for image processing applications. CNNs are capable of automatically learning critical features from images, which are then used to identify and classify the presence of specific objects in a driving scenario.

In this research, the YOLO model, which is a CNN-based model, will be trained using synthetic datasets created using simulation environments such as CARLA. The datasets created using such tools will be labeled images of road scenarios. Training the YOLO model using such datasets will enable the system to learn how to effectively identify objects in a scenario while also assessing the impact of data generation in a privacy-preserving fashion.

Additionally, the process of training the model will be carried out in a structured pipeline, which includes data preprocessing, splitting the data into training, validation, and test sets, and monitoring the performance of the model during training. This ensures consistency and reliability in the evaluation of the model's performance, whether the data is real or synthetic.

3.3 Tools and Platforms

Table 3.3: Tools and Platforms

Tool	Purpose
CARLA Simulator	Synthetic dataset generation
PyTorch	Deep learning model training
YOLO	Object detection model
Python	Data processing and model development
KITTI Dataset	Baseline evaluation dataset

3.4 Data Collection and Ethical Considerations

The synthetic data would be created using the CARLA simulator. Since synthetic data does not contain personal information, there are no privacy issues.

The ethical considerations are:

- ➔ There should be no exposure of real world personal data.
- ➔ There should be transparency and maintaining transparency in dataset generation.
- ➔ Compliance with data privacy regulations like GDPR.

3.5 Validation Strategy and Performance Metrics

To assess how well the trained perception models perform, the models will be validated with a set of data known as the validation data set. The performance of the perception models will be measured using various performance metrics for object detection models.

The performance metrics for this particular study will be Mean Average Precision (mAP), Precision, Recall and F1 Score metrics.

Besides, Frechet Inception Distance (FID) will be employed to assess the quality and realism of the generated synthetic datasets by comparing them with the distribution of the real world datasets.

The performance of the model will be assessed by comparing the results of the following sources:

- Real – world datasets (baseline)
- Privacy – preserving synthetic datasets

This allows for a clear assessment of the impact of the privacy constraint on the quality of the data as well as the performance of the model.

The mAP metric is used to measure the overall precision of the object detection model. This metric shows how well the model performs in detecting and recognizing objects such as vehicles, pedestrians and signs.

Precision is used to measure how many of the detected objects by the model are actually correct. Precision is high when hew incorrect detections are made by the model.

The concept of Recall measures the number of actual objects in the image that are successfully identified by the model. A higher value of recall indicates that the model is able to identify most of the objects in the scene.

F1 Score is a measure that combines both the precision and recall of the model. This score can be used to evaluate the performance of the detection model.

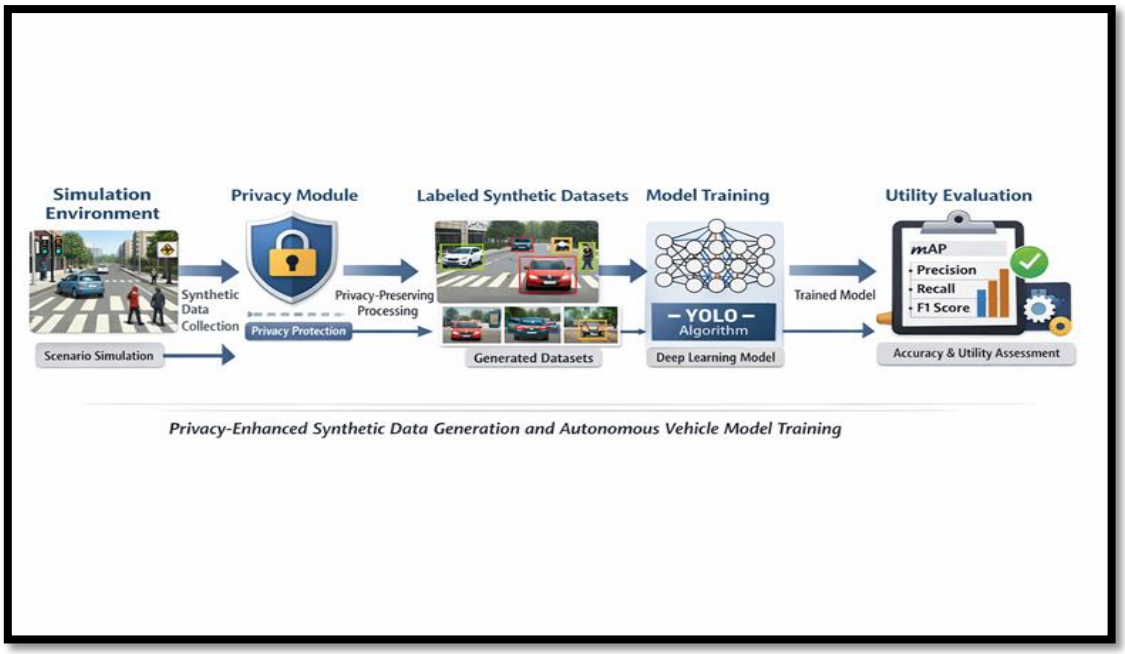
The above metrics for evaluating the performance of the perception models are to be used for the research to investigate the effect of privacy constraints on the performance of the perception models. By comparing the results obtained under different privacy settings, the research aims to evaluate the privacy utility trade off and check whether synthetic data can be of use for training autonomous vehicle perception models.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and its Components

The proposed system is intended for generating privacy enhanced synthetic datasets and evaluating the usefulness of these datasets for training autonomous vehicle perception models. The proposed system comprises different components that cooperate to achieve the intended functionality. These components are:

- Firstly, the proposed system simulates different driving conditions using a simulation environment. In the process, different objects such as cars, pedestrians, traffic signs, and different road conditions are included.
- After simulating different conditions, synthetic images are collected.
- Secondly, the proposed system applies privacy constraints using the privacy module. In the process, privacy-preserving mechanisms are employed during the generation process.
- After applying the privacy constraints, the proposed system generates labeled synthetic datasets.
- Lastly, the proposed system uses the generated datasets for training deep learning models using the model training component. In the process, deep learning models such as YOLO are used for object detection.
- There is a utility evaluation module that will evaluate the efficiency of the trained models based on standard metrics. This module will examine the impact of privacy constraints on model accuracy and system efficiency.
- Generally, this system architecture is essential in allowing researchers to examine the link between privacy protection and model efficiency, which is the aim of this research.



4.2 Main Components of the System

1. Simulation Environment: -

- This is the simulation environment module. This module creates virtual driving scenarios. The simulation is usually carried out using a simulator like CARLA. The simulator is used to create scenarios with vehicles, pedestrians, traffic lights, signs, and various weather conditions.
- The simulator captures images from the virtual environment. The images are then considered raw data to create synthetic data.

2. Privacy Module: -

- This is the privacy module. The privacy module is responsible for ensuring privacy preserving techniques are applied during the generation of the data sets. The techniques are essential in ensuring that sensitive data is not exposed.
- Ex: - there are privacy constraints that can change images to ensure that sensitive data is not exposed.

3. Dataset Generation Module: -

- The dataset generation module is used to transform the data obtained from the simulator into datasets that can be used for machine learning.
- This module is used to retrieve images from the simulator's simulation environment and generate labels for vehicles, pedestrians, and signs in the images.
- The images with labels are then used to create training datasets for use in the perception model training.

4. Model Training Module: -

- The model training module is used to train perception models for autonomous vehicles using synthetic datasets.
- Deep learning techniques like YOLO object detection using Convolutional Neural Networks (CNN) are used in this module for training perception models for vehicles.
- This component is used to evaluate whether synthetic datasets generated using privacy – preserving techniques can be used for model training.
- The model training module will also include the development of a complete pipeline, such as data preprocessing, data splitting, hyper parameter tuning, and tracking. The models trained on synthetic data will be compared with baseline models trained on actual data in terms of effectiveness.

5. Utility Evaluation Module: -

- The purpose of the utility evaluation module is to evaluate the performance of the trained perception models.
- The performance of the models can be evaluated with the use of the following object detection performance metrics:
  - ➔ Mean Average Precision (mAP)
  - ➔ Precision
  - ➔ Recall
  - ➔ F1 Score
- The performance metrics will be used to evaluate the accuracy and reliability of the trained models with the use of synthetic data sets.

4.3 Data Flow Explanation

The data flow of the system is composed of a series of sequential processes that are necessary in order to complete a task / a series of tasks.

- The following is a description of the processes involved in the creation of synthetic datasets with enhanced privacy features, as well as the evaluation of the effectiveness of the synthetic datasets for training autonomous vehicles' perception models. The processes are shown in the following data flow diagram:

1. Simulator: -
   - ➔ The first step involves the use of a simulation environment, which is a driving simulator. The simulator is used to create a virtual environment that represents realistic driving situations, which may include vehicles, pedestrians, traffic signs and road conditions. The images used in the simulation represent synthetic sensor data, which may include images of different situations, for instance, vehicles, pedestrians etc.

➔ This stage is responsible for producing the initial data required for the dataset generation process.

2. Synthetic Dataset: -
   ➔ The data collected from the simulator is further processed to create a structured synthetic dataset. The images taken from the simulation environment are collected for machine learning training.
   ➔ During the process, object annotations such as bounding boxes and labels are performed to identify objects such as vehicles, pedestrians and traffic signs. The images obtained from the simulator are the synthetic dataset for training perception algorithms.
   ➔ In the expanded project plan, this process also involves the automatic extraction of ground truth labels for the data collected.

3. Model Training: -
   ➔ The synthetic dataset is then used to train the perception models for an autonomous vehicle. For instance, deep learning-based models such as the YOLO model can be trained for object detection in the images created in the synthetic dataset.
   ➔ The training process involves the learning of patterns from the images in the dataset and the adjustment of parameters in the model to optimize the results. The training process is one of the main tasks that can be carried out by me.
   ➔ During this stage, the dataset is divided into training, validation and testing subsets to ensure proper evaluation. The training process includes configuring model parameters and monitoring performance across epochs.

4. Performance Evaluation: -
   ➔ After training the model, the next step is to evaluate the model's performance using a series of tests to determine how well the model can identify objects within the dataset.
   ➔ We usually check the following metrics:
      - Mean Average Precision (mAP)
      - Precision
      - Recall
      - F1 Score

   ➔ With these metrics, we can get a good idea of how reliable the model's results are.
   ➔ Evaluating the metrics is one of the main responsibility of me, as the main goal is to ensure that the synthetic dataset with enhanced privacy features can be used for training autonomous.

5. Trade – off Analysis: -
   - ➔ In the last step of the data flow, we are concerned with how the privacy features affect the overall performance of the model. This is because the privacy features are embedded within the generation of the data, so it is essential to determine how the constraints affect the overall accuracy of the models that are developed.
   - ➔ In the trade-off analysis, the results are analyzed to determine whether the enhanced privacy features result in a trade-off with the overall performance of the models. This is achieved by using graphs to compare the level of privacy with the overall accuracy of the models.
   - ➔ The analysis is focused on achieving a balance between the data privacy of the data and the overall effectiveness of the machine learning models, which is the primary goal of the project and a major objective for me.

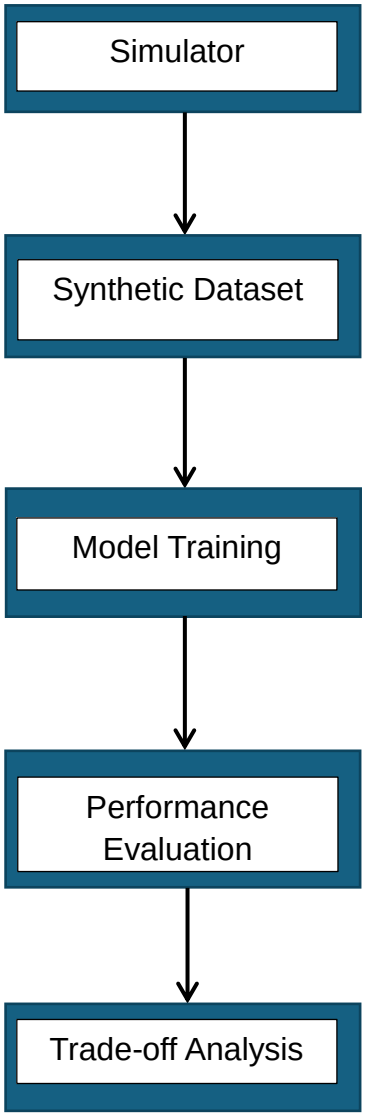


Figure 4.3: Data Flow Explanation

- The simulator generates virtual driving environments and captures images.
- These images are processed to create a synthetic dataset with object labels.
- The dataset is used to train the perception models.
- The trained models are then tested for their performance.
- Finally, the results are analyzed to determine the privacy-utility trade-off.

4.4 Scalability and Security Considerations

4.4.1. Cloud Storage for Large Datasets: -

- The synthetic dataset generation process has the capability to generate large volumes of image data. Thus, cloud storage can be utilized to manage the data effectively.

4.4.2. Secure Model Training Pipeline: -

- The model training process must be secured in a secure computing environment to ensure the dataset and the models are not compromised or accessed improperly.

4.4.3. Privacy – Aware Dataset Generation: -

- The integration of the dataset generation process with the aspect of data privacy has been considered. This will ensure that the generated data follows the principles of data privacy, which will then guarantee the data's usefulness.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- Stakeholders are individuals or groups that utilize the system or benefit in some way.

5.1.1 Autonomous Vehicle Researchers:

- These are the individuals who use this system to investigate the impact of privacy – preserving synthetic data sets on training and using autonomous vehicle perception systems.

5.1.2 AI Developers:

- These are individuals who use this system to develop and enhance object detection algorithms, which are part of the autonomous driving system.

5.1.3 Autonomous Companies:

- This group can use this system to develop and enhance autonomous vehicle systems while ensuring privacy.

5.1.4 Regulatory Authorities:

- This group can use this system to check whether the generation of data sets is privacy compliant and can be used responsibly in developing and using AI in autonomous vehicles.

5.2 Functional Requirements

- Functional requirements are those that describe what the system should do.

5.2.1 The system shall generate synthetic driving datasets.

- The system must generate realistic driving data through a simulation environment that includes vehicles, pedestrians and scenarios on the road.

5.2.2 The system shall enforce privacy mechanisms during dataset generation.

- The system must enforce privacy constraints during dataset generation.

5.2.3 The system shall train AV perception models using generated datasets.

- The system must use the generated datasets to train deep learning models to detect objects within an autonomous driving environment.

5.2.4 The system shall evaluate model performance.

- The system must evaluate the performance of the models through metrics such as mAP, precision, recall and F1 score.

5.2.5 The system shall analyze privacy – utility trade-offs.

- The system must evaluate the impact of enforcing privacy constraints on the accuracy of the perception models.

5.3 Non – Functional Requirements

- Non - functional requirements are requirements that state how well the system should perform.

5.3.1 High computational performance.

- The system should be able to process data and train perception models using high computational performance.

5.3.2 Data privacy protection.

- The system must be able to ensure the use of privacy protection mechanisms during the generation of synthetic data.

5.3.3 Scalability for large datasets.

- The system should be able to handle and process synthetic data in large quantities without compromising performance.

5.3.4 Model accuracy.

- The perception models must be able to maintain high accuracy even if they are trained using synthetic data generated by privacy protection mechanisms.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed system may be of interest to a number of organizations that are engaged in the development and use of autonomous vehicles and artificial intelligence.

Autonomous Vehicle Manufacturers:

Autonomous vehicle manufacturers require a lot of data to train the perception system. The proposed system might be able to offer a way of generating synthetic data, which might be used to train the artificial intelligence models without the manufacturer needing any sensitive data.

AI Research Institutions:

Research institutions and universities engaged in the development of artificial intelligence may be able to benefit from the proposed system in the development of autonomous vehicles.

Smart Mobility Companies:

Companies engaged in the development of smart mobility solutions such as intelligent traffic systems may benefit from the proposed system by being able to generate large amounts of data to train their machine learning models.

6.2 Value Proposition

The value of the proposed system lies in the fact that it offers a private synthetic dataset that can be used in the development of an autonomous vehicle perception system.

In various real-life cases, using a driving dataset may prove problematic in terms of ensuring the privacy of the data used. For instance, the presence of personal information in the dataset may cause problems. However, the proposed system ensures the creation of a synthetic dataset with the incorporation of privacy constraints.

Additionally, the proposed system ensures a trade-off in terms of privacy and utility. This means that organizations can use the system in developing an AI system with the incorporation of privacy.

6.3 Revenue Model

The system has various ways to earn revenue through different business models.

Dataset Licensing:

Organizations can be given license to access and use the generated synthetic datasets for training AI models.

Simulation Platform Integration:

The system can be integrated into existing simulation platforms used by developers of autonomous vehicles, hence providing organizations with the ability to generate privacy – aware datasets within their own environment.

Research Collaboration:

The system can be used to collaborate with universities to carry out research projects on privacy-aware AI and autonomous vehicles.

6.4 Competitive Advantage

The proposed system has several advantages over existing approaches for generating datasets.

Privacy-Aware Dataset Generation:

The system includes the direct application of privacy – preserving methods during the dataset generation process.

Quantified Privacy-Utility Trade-off:

This research is unique from existing approaches as it quantifies the impact of privacy preservation on model accuracy. This is very useful for organizations that need a delicate balance between data privacy and AI model accuracy.

Reduced dependency on real-world datasets:

The cost of obtaining driving data in the real world is extremely expensive and heavily regulated by data privacy laws.

6.5 Cost Estimation

The various costs associated with the development of the system and its maintenance includes computing resources, data storage and development of the system.

High – performance computing resources might be necessary for simulations, generation of datasets and development of models. Cloud storage might also be necessary for the storage of large amounts of synthetically generated datasets.

The additional costs might include maintenance of the system, software development, and research activities associated with improving the process of dataset generation and evaluation.

6.6 Pricing Strategy

Organizations that run dataset generation repeatedly can pay a recurring subscription, with fees scaled to compute volume and configuration. Teams with occasional need pay per certified run instead. Simulator vendors, such as those building on CARLA or NVIDIA DRIVE Sim, can license the ε-DP pipeline directly and ship it as part of their own platform under an annual agreement.

6.7 Intellectual Property Considerations

The results of the research, including the data set generation pipeline, the methods of enforcing privacy and the evaluation process, may be subject to intellectual property rights.

The software used in the project may be copyrighted / patented in order to safeguard the innovations in the project. This ensures that the technology developed in the research process can be used in a commercial environment.

7. BUDGET AND JUSTIFICATION

- The aim of this section is to estimate the resources needed in implementing the proposed research system. It should be noted that the research project mainly entails training deep learning models, developing synthetic datasets and assessing the performance of the models. Therefore, the budget will include costs related to hardware use, cloud storage, software tools and research.

Table 7: Budget Estimation for Research Project

Category	Item	Description	Estimated Cost (LKR)
Hardware Costs	CPU computational resources	Existing laptop CPU used for deep learning model training and experimentation.	0
Software Licenses	Development Tools	Most tools are open-source (Python, PyTorch, YOLO, CARLA). Small cost allocated for additional utilities if needed.	2,000
Cloud Services	Cloud Storage	Storage required for saving synthetic datasets and trained model outputs.	5,000
Human Effort Cost	Research Development Effort	Research development, model training and system evaluation work.	5,000
Miscellaneous	Additional Research Expenses	Internet usage, backup storage devices and unexpected research costs.	2,000
Total Estimated Budget: 14,000 LKR			

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown per Member

Tasks are assigned to members according to their individual research focus. Phases, weeks and key tasks for all four members are listed on the table.

Table 8.1: Work Breakdown Structure (WBS)

Member	Role	Phase	Weeks	Key Tasks
IT22309556	Privacy Integration	1	1 – 3	Literature review on DP mechanisms. Library selection (Opacus vs TF Privacy). Define ε range for ablation
IT22309556	Privacy Integration	2	4 – 7	Build DP training wrapper. Implement Privacy – Aware Calibration Module with Laplace noise. Run ablation experiments.
IT22309556	Privacy Integration	3	8 – 10	Full – scale generation run under ε-DP. Monitor ε consumption. Confirm budget not exceeded.
IT22309556	Privacy Integration	4	11 – 14	Coordinate with IT22066916 on MIA tests.
IT22541284	Dataset Generation / Simulation	1	1 – 3	Set up CARLA simulator environment. Define scene and sensor configurations.
IT22541284	Dataset Generation / Simulation	2	4 – 8	Run CARLA simulation using privatized calibration statistics from IT22309556. Generate raw synthetic dataset batches.
IT22541284	Dataset Generation / Simulation	3	9 – 12	Validate dataset format and sensor coverage. Deliver final synthetic dataset to

				IT22110220.
IT22110220	Utility Evaluation	1	1 – 4	Define utility metrics (mAP, FID). Set up AV model training pipeline and baseline benchmarks.
IT22110220	Utility Evaluation	2	8 – 11	Train AV model on synthetic dataset. Measure mAP and FID against non – private baseline.
IT22110220	Utility Evaluation	3	12 – 14	Report utility trade – off results. Contribute utility analysis to group thesis.
IT22066916	Risk and Compliance	1	1 – 3	Review GDPR, CCPA and Sri Lanka PDPA requirements. Define compliance checklist.
IT22066916	Risk and Compliance	2	9 – 12	Run membership inference and re – identification attack tests on DP synthetic data from IT22309556.
IT22066916	Risk and Compliance	3	12 – 14	Compile attack results and compliance report. Contribute risk and regulatory sections to group thesis.

8.2 Timeline Alignment

Shaded cells mark weeks where a member has active tasks. Phases overlap at inter-member handoff points: IT22309556 delivers privatized calibration statistics to IT22541284 at the end of Phase 1 and delivers the ε-DP audit log to IT22066916 at the start of Phase 3.

Table 8.2: Timeline Alignment

Member	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
IT22309556	1	2	3	4	5	6	7	8	9	10	11	12	13	14
IT22541284	1	2	3	4	5	6	7	8	9	10	11	12		
IT22110220	1	2	3	4				8	9	10	11	12	13	14
IT22066916	1	2	3						9	10	11	12	13	14

8.3 Balanced Workload Distribution

Active week counts run from 9 to 14 across the four members. IT22309556 covers all 14 weeks because the budget requires continuous monitoring throughput every phase. IT22541284 and IT22110220 each run 11 and 12 active weeks. IT22066916 covers 9 weeks, concentrated in the attack testing and compliance reporting phase. The difference in week counts follows from the nature of each role: privacy integration runs end-to-end, while utility evaluation and compliance testing fall within bounded phases.

Table 8.3: Balanced Workload Distribution

Member	Active Weeks	Phases	Primary Deliverables
IT22309556	1 – 14 (14 weeks)	4	ε-DP training wrapper, calibration module, ablation results , privacy budget log.
IT22541284	1 – 12 (12 weeks)	4	CARLA simulation setup, synthetic dataset batches, validated dataset delivery.
IT22110220	1 – 4 , 8 – 14 (11 weeks)	4	Utility metric framework, AV model training results, mAP and FID analysis report.
IT22066916	1 – 3 , 9 – 14 (9 weeks)	4	Compliance checklist, MIA and re – identification attack results, regulatory compliance report.

9. GANTT CHART

Task	Semester 1 (Mar – May 2026)			Semester 2 (Jun – Dec 2026)						
All Members	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Literature Deep Dive and Architecture Mapping										
Hardware and Storage Allocation Setup										
Calibration Strategy and Integration Checkpoint										
Full Pipeline Execution and QA Review										
Metric Compilation and Final Paper Drafting										
SLIIT ASSESSMENT MILESTONES										
★ Proposal Submission (15 Mar 2026)	★									
★ Proposal Presentation (16–18 Mar 2026)	★									
★ Progress Presentation 1 (4–6 May 2026)			★							
★ Progress Presentation 2 (17–19 Aug 2026)						★				
★ Final Report Submission (11 Oct 2026)								★		
★ Final Presentation and Viva (12–14 Oct 2026)								★		
★ Research Paper Submission (15 Oct 2026)								★		
★ Publication Evidence Submission (1 Dec 2026)										★

Figure 9: Gantt Chart

- Star cells mark SLIIT official milestones. Semester 1 runs from March to May 2026; Semester 2 runs from June to December 2026.

10.REFERENCES LIST

[1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 779–788, 2016.

[2] J. Redmon and A. Farhadi, "YOLOv3: An Incremental Improvement," *arXiv preprint arXiv:1804.02767*, 2018.

[3] A. Dosovitskiy *et al.*, "CARLA: An Open Urban Driving Simulator," *Proceedings of the Conference on Robot Learning (CoRL)*, pp. 1–16, 2017.

[4] S. R. Richter, V. Vineet, S. Roth, and V. Koltun, "Playing for Data: Ground Truth from Computer Games," *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 102–118, 2016.

[5] C. Dwork and A. Roth, "The Algorithmic Foundations of Differential Privacy," *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3–4, pp. 211–407, 2014.

[6] R. Shokri *et al.*, "Membership Inference Attacks Against Machine Learning Models," *Proceedings of the IEEE Symposium on Security and Privacy*, pp. 3–18, 2017.

[7] A. Geiger, P. Lenz, and R. Urtasun, "Are We Ready for Autonomous Driving? The KITTI Vision Benchmark Suite," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3354–3361, 2012.

[8] H. Caesar *et al.*, "nuScenes: A Multimodal Dataset for Autonomous Driving," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11621–11631, 2020.

[9] M. Everingham *et al.*, "The Pascal Visual Object Classes (VOC) Challenge," *International Journal of Computer Vision*, vol. 88, no. 2, pp. 303–338, 2010.

[10] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

[11] W. Liu *et al.*, "SSD: Single Shot MultiBox Detector," *European Conference on Computer Vision (ECCV)*, pp. 21–37, 2016.

[12] A. Kendall and Y. Gal, "What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?" *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

11.APPENDICES

11.1 Appendix A – Survey Instruments

Table 11.1: Appendix A – Survey Instruments

Component	Description
Purpose of Survey	To collect feedback from AI researchers, autonomous vehicle developers, and data scientists about the usefulness of privacy-preserving synthetic datasets.
Target Participants	Autonomous vehicle researchers, AI developers, and data scientists working with machine learning datasets.
Key Survey Topic 1	Importance of synthetic datasets for training autonomous vehicle perception models.
Key Survey Topic 2	Challenges faced when using real-world datasets due to privacy concerns.
Key Survey Topic 3	Impact of privacy constraints on the performance of machine learning models.
Key Survey Topic 4	Acceptance of synthetic datasets as an alternative to real-world driving data.
Key Survey Topic 5	Evaluation metrics used to assess model performance such as mAP, Precision, Recall, and F1 Score.
Relation to IT22110220	Helps validate the effectiveness of privacy-preserving datasets and supports the privacy–utility trade-off analysis conducted in the research.

11.2 Appendix B – Detailed System Diagrams

Table 11.2: Appendix B – Detailed System Diagrams

Diagram Type	Description
Overall System Architecture Diagram	Shows the interaction between the simulation environment, privacy enforcement module, dataset generation module, model training module, and utility evaluation module.
Data Flow Diagram	Illustrates how simulated data flows from the simulator to dataset generation, model training, and performance evaluation stages.
Model Evaluation Workflow Diagram	Demonstrates the process of training perception models and evaluating them using metrics such as Mean Average Precision (mAP), Precision, Recall, and F1 Score.
Relation to IT22110220	Highlights the model training, performance evaluation, and privacy–utility analysis responsibilities assigned to IT22110220. Such as AV model training pipeline setup, baseline benchmarking using real – world datasets, performance evaluation using mAP, Precision, Recall, F1 Score, and FID metrics and finally carrying out privacy – utility trade – off analysis.

11.3 Appendix C – Preliminary User Interface Mockups

Table 11.3: Appendix C – Preliminary User Interface Mockups

UI Component	Description
Dataset Overview Panel	Displays information about generated synthetic datasets including number of images, object classes, and annotations.
Training Progress Dashboard	Shows training progress of the deep learning models including training loss curves and training parameters.
Performance Metrics Visualization Panel	Displays evaluation results using metrics such as mAP, precision, recall, and F1 score.
Privacy–Utility Trade-off Graph	Graphically represents the relationship between privacy constraints and model performance.
Purpose	Allows researchers to easily monitor dataset generation, model training, and evaluation results.

11.4 Appendix D – Risk Analysis

Table 11.4: Appendix D – Risk Analysis

Risk	Description	Mitigation Strategy
High Computational Requirements	Deep learning model training may require significant computational resources which can increase training time when using CPU resources.	Optimize training parameters and reduce dataset size for experiments when necessary.
Synthetic Data Realism Limitations	Synthetic datasets may not perfectly represent complex real-world driving scenarios.	Improve simulation parameters and use diverse simulation environments.
Privacy Parameter Misconfiguration	Incorrect privacy settings may reduce model accuracy or fail to protect sensitive information.	Carefully tune privacy parameters and validate results during testing.
Dataset Imbalance	Generated datasets may contain unequal object distributions.	Apply dataset balancing techniques during dataset generation.
Security Risks	Attackers may attempt to perform inference attacks to extract information from trained models.	Implement privacy-preserving techniques and security testing procedures.

11.5 Appendix E – Ethical Documents

Table 11.5: Appendix E – Ethical Documents

Ethical Aspect	Description
Data Privacy Protection	The system generates synthetic datasets instead of using identifiable real-world personal data.
Responsible AI Development	The research ensures that AI models are trained using ethically generated datasets.
Regulatory Compliance	The system follows privacy-by-design principles aligned with data protection regulations such as GDPR.
Transparency	The research clearly reports model evaluation metrics and experimental results to ensure transparency.
Research Integrity	The project follows academic research guidelines and ethical standards during data generation and experimentation.